Q.1] Print an n×n star Square.

$\boxed{n=4}$

```
~~import~~
import java.util.Scanner;
class Demo {
  Public static void main (String args[]) {
    Scanner scan = new Scanner (System.in);
    sop ("Enter value of n:-");
      int n = Scan.nextInt();
    for (int i = 1 ; i<=n ; i++) {
        for (int j = 1 ; j<=n ; j++) {
          sop ("*");
        }
      sopln();
    }
  }
}
```

Dryrun n=4

i=1

1≤4 j<n

j=1 j≤4 j++

*

j=2 2≤4 j++

**

j=3 3≤4 j++

***

j=4 4≤4 j++

****

j=5 5≤4 j++

false

i=2

2≤4

j=1 j≤4 j++

****

*

j=2 2≤4 j++

****

**

j=3 3≤4 j++

****

***

j=4 3≤4 j++

****

****

j=5 5≤4 j++

false

Output

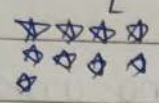
```
* * * *
* * * *
* * * *
* * * *
```

Page No.
Date: / /

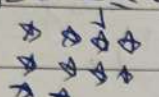
$i = 3$

$3 \leq 4$

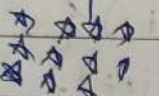
$j=1$ $1 \leq 4$; j++



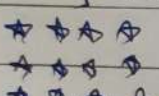
$j=2$ $2 \leq 4$; j++



$j=3$ $3 \leq 4$ j++



$j=4$ $4 \leq 4$ j++



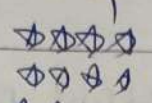
$j=5$ $5 \leq 4$ j++

False

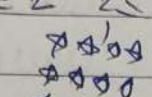
$i=4$

$4 \leq 4$

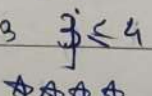
$j=1$ $1 \leq 4$; j++



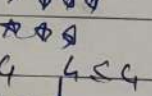
$j=2$ $2 \leq 4$; j++



$j=3$ $3 \leq 4$; j++



$j=4$ $4 \leq 4$; j++



$j=5$ $5 \leq 4$; j++

False

$i=5$

$5 \leq 5$

False.

1) What does outer loop represent

→

e) What does inner loop represent

→

Q.3] Print the following Pattern n=4

```
-> import java.util.*;
   class Demo {
    Public Static void main (string args[]){
    Scanner scan = new scanner (system.in);
    SOP ("Enter value of n:");
int n=Scan.nextInt();
      for (int i=1 ; i<=n ; i++){
          for (int j=1 ; i<=n ; j++){
              SOP (j+ "_");
              }
          sopln();
      }}}
```

Output:- **Prgrm** n=4

```
1 2 3 4     i=1                 i=2
1 2 3 4     1<=4                2<=2
1 2 3 4     j=1 j<=4 j++        j=1 1<=4 j++
1 2 3 4     1                   1234
            j=2 2<=4 j++        j=1 2<=4 j++
            12                  1234
                                12
            j=3 3<=3 j++        j=3 3<=4 j++
            123                 1234
                                123
            j=4 4<=4 j++        j=4 4<=4 j++
            1234                1234
                                1234
            j=5 5<=4 j++        j=5 5<=4 j++
            False               False
```

Page No.
Date.: / /

i = 3

3 ≤ 4

j=1 1 ≤ 4 j++
1234
1234
1

j=2 2 ≤ 4 j++
1234
1234
12

j=3 3 ≤ 4 j++
1234
1234
123

j=4 4 ≤ 4 j++
1234
1234
1234

j=5 5 ≤ 4
false

i=4

4 ≤ 4

j=1 1 ≤ 4 j++
1234
1234
1234
~~1234~~

j=2 2 ≤ 4 j++
1234
1234
1234
12

j=3 3 ≤ 4 j++
1234
1234
1234
123

j=4 4 ≤ 4 j++
1234
1234
1234
1234

j=4 5 ≤ 4
false

i = 5

5 ≤ 4
false

Q.3] Print the following pattern n=4

```
import java.util.*;
class Demo {
public static void main (String args[]) {
Scanner scan = new Scanner (System.in);
SOP ("Enter value of n:");
int n = scan.nextInt();
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; j≤n ; j++) {
        SOP (i+" ");
    }
    SOPln();
}
}
```

Dry run n=4

i=1 1≤4

j=1 1≤4 j++
1

j=2 2≤4 j++
1 1

j=3 3≤4 j++
1 1 1

j=4 4≤4 j++
1 1 1 1

j=5 5≤4 j++
false

i=2 2≤4

j=1 1≤4 j++
1 1 1 1
2 2 2

j=2 2≤4 j++
1 1 1 1
2 2

j=3 3≤4 j++
1 1 1 1
2 2 2

j=4 4≤4 j++
1 1 1 1
2 2 2 2

j=5 5≤4 j++
false

Output
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

i=3 3≤4

j=1 1≤4 j++ (i)

1111
2222
3

j=2 2≤4 j++

1111
2222
33

j=3 3≤4 j++

1111
2222
333

j=4 4≤4 j++

1111
2222
3333

j=5 5≤4 j++

false

i=4 4≤4

j=1 1≤4 j++ (i)

1111
2222
3333
4

j=2 2≤4 j++

1111
2222
3333
44

j=3 3≤4 j++

1111
2222
3333
444

j=4 4≤4 j++

1111
2222
3333
4444

j=5 5≤4 j++

false

i=5 5≤4

false

Q.i